

FinTech Project 2026

Systematic Multi-Asset Funds with News-Sentiment Analytics

FINS5545: Financial Market Data Literacy

The product you will build

- You will build your own proto-type for an investment app – e.g., UNSW-Trader
- The app comprises systematically managed funds, each a portfolio of stocks and crypto built with rules-based portfolio optimizations
- e.g., An investor opens the app, compares the funds, and reads a short fact sheet for each one.
- They then decide how to split their money across the funds.
- The funds are the 'product the app sells', and the business (your app) earns a management fee on the money people invest.

From raw data to investable funds

- The funds come from market prices — daily prices become returns, and returns become optimal portfolios –i.e., the Data Factory Floor
- An optimal portfolio is, for example, the blend of assets that gives the best return for the risk taken.
- You test each fund out of sample, so the performance you report is what it would have earned on data it had not already seen.
- Alongside the prices, you score news headlines into a sentiment index for each equity sector that a user can follow over time.
- The richest part of the project is bringing the two together — letting sentiment tilt the portfolio and judging whether it helps (or not)?
- Innovation has a very high weighting

The Data Factory Floor

The project follows the four stages of the Data Factory Floor, the path data takes from raw files to a finished product.

- Station 1 — collect and clean the data.
- Station 2 — build the return features and assemble the news text.
- Station 3 — design the models, both the portfolios and the sentiment.
- Station 4 — deliver it all as a working app.

Part A is the first two stations, the data foundation. **Part B** is the last two, where the funds, the sentiment, and the app come together.

How the project is graded

- Part A is worth **20%** of your course mark and Part B **50%**, so the project carries most of the course.
- You may use AI for all of it, and we expect you to.
- AI can produce a competent baseline on its own, so a large proportion of the grades come from:
 - original ideas that go beyond the baseline
 - your own reading of what the results mean
 - a clear record of how you worked with the AI tools and where you corrected it
 - generating truly novel results/insights
- A project that only prompts and pastes will not do very well

What you hand in

- Two short written reports and a live app, deployed to the web from your own code.
- All the data is provided — fifty US stocks across ten sectors, ten cryptocurrencies, and four years of news headlines — so there is nothing to scrape or download.
- The stock data has both prices and news, so the sentiment work is only on the equity side.
- Crypto comes in as a price-only diversifier (no unstructured data here)
- Everything you design will be completed in one project folder, and that folder is what you run, what you submit, and what becomes your deployed app – all in `fins-agent`